UA-66 Urine Analyzer

Operator’s Manual

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- the electrical installation of the relevant room complies with the applicable national and local requirements; and
- the product is used in accordance with the instructions for use.

WARNING:

It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine breakdown or personal injury.

NOTE:

This equipment must be operated by skilled/trained clinical professionals.

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- Malfunction of the instrument or part whose serial number is not legible enough.
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Preface

Safety Symbols and Labels

Labels are used to indicate potential hazards in operating the system. The following table explains the labels used on the system.

	Electric shock: Reminding users of electric shock. Note: Do not remove the shell. Otherwise, electric shock may be caused.
	Warning: Reminding users to operate as instructed in this manual. Otherwise, personal injury may be caused.
	Biohazard: Reminding users to operate as instructed in this manual to prevent biohazardous risks.
SN	Serial number
	Date of manufacture
	Manufacturer
IVD	In Vitro Diagnostic equipment
	Storage temperature
	Atmospheric pressure range for storage
	Relative humidity for storage

Safety Precautions

Observe the following safety precautions when using the product. Ignoring any of these safety precautions may lead to personal injury or equipment damage.

Warning:

If the product is used in a manner not specified by our company, the protection provided by the product may be impaired.

Biohazard

Observe the following instructions to protect against biohazardous risks.

Biohazard:

Inappropriately handling samples may lead to biohazardous infection. Do not touch samples with your bare hands. Wear gloves and lab coat and, if necessary, goggles.

In case your skin contacts samples and mixtures, follow the standard laboratory safety procedure and consult a doctor.

Waste Disposal

Observe the following instructions to prevent environmental pollution and personal injury caused by waste.

Biohazard:

Some substances contained in the whole device and the parts are subject to regulations of contamination and disposal. Dispose of the waste in accordance with your local or national rule for biohazard waste disposal. Wear gloves and lab coat and, if necessary, goggles.

Electromagnetic Noise Precautions

Caution:

This equipment complies with the emission and immunity requirements of the IEC 61326-1:2005 and IEC 61326-2-6:2005.

This equipment has been designed and tested to CISPR 11 Class A. In a domestic environment it may cause radio interference, in which case, you may need to take measures to mitigate the interference.

Do not use this device in close proximity to sources of strong electromagnetic radiation (e.g. unshielded intentional RF sources), as these may interfere with the proper operation.

Note:

It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.
It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that the device will perform as intended.

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1 System Description

1.1 Intended Use

This device is intended for routine determination of the following clinical test items in urine: Glucose (GLU), Bilirubin (BIL), Specific Gravity (SG), PH, Ketone (KET), Blood (BLD), Protein (PRO), Urobilinogen (URO), Nitrite (NIT), Leukocytes (LEU), and Ascorbic Acid (VC).

1.2 Technical Specifications

Specification	Description
Test items:	GLU, BIL, SG, PH, KET, BLD, PRO, URO, NIT, LEU, and VC
Working mode:	single step mode 、 fast batch mode
Throughput:	60 tests per hour(single step mode)、120 tests per hour(fast batch mode)
Wavelength:	Red (R), Green (G), and Blue (B)
Language:	English
Recording mode:	Micro thermal printer, LCD, FalshROM, and external printer
Display monitor:	LCD, 240*128

Specification	Description
Print:	Micro thermal print paper, width of 57mm, roll diameter less than 50mm External printer
Communication interface	RS-232C DB9 female
Operating temperature and relative humidity:	10°C-30°C, ≤80%
Atmospheric pressure:	76kPa~106kPa
Transportation and storage conditions:	Temperature; 0°C~40°C Relative humidity: ≤80%
Operating environment:	The device should be placed on a stable plane platform, kept from direct sunshine and strong electromagnetic interference, and properly grounded.
Power input:	AC100-240V, 50/60Hz Main supply voltage fluctuations up to ±10% of the nominal voltage.
Fuse:	ф5mm*20mm, 250V/2A
Power consumption:	40VA
Safety classification:	Electric shock protection I type, overvoltage type class II, pollution degree 2
Dimensions:	362mm×303mm×125mm

Table 1.1

1.3 EMC Description

- Do not use this device in close proximity to sources of strong electromagnetic radiation (e.g. unshielded intentional RF sources), as these may interfere with the proper operation.

- This equipment complies with the emission and immunity requirements of the EN 61326-1:2006 and EN 61326-2-6:2006.

  This equipment has been designed and tested to CISPR 11 Class A. In a domestic environment it may cause radio interference, in which case, you may need to take measures to mitigate the interference.

Note:

It is the manufacturer's responsibility to provide equipment electromagnetic compatibility information to the customer or user.

It is the user's responsibility to ensure that a compatible electromagnetic environment for the equipment can be maintained in order that the device will perform as intended.

2 Result Units

Test Item	Arbitrary	Conventional Units	International Units
LEU	-	neg.	
	±	15 cells/μL	
	1+	70 cells/μL	
	2+	125cells /μL	
	3+	500cells/μL	
URO	norm.	norm.	norm.
	1+	2.0 mg/dL	33 μmol/L
	2+	4.0 mg/dL	66 μmol/L
	3+	8.0 mg/dL	131 μmol/L
	4+	12.0 mg/dL	200 μmol/L
BLD	-	neg.	neg.
	1+	0.06 mg/dL	25 cells/μL
	2+	0.15 mg/dL	70 cells/μL
	3+	0.75 mg/dL	250cells/μL
BIL	-	neg.	Neg.
	1+	0.5 mg/dL	8.6 μmol/L
	2+	1.0 mg/dL	17 μmol/L
	3+	2.0 mg/dL	50 μmol/L

Result Units

Test Item	Arbitrary	Conventional Units	International Units
KET	-	neg.	neg.
	1+	15 mg/dL	1.5 mmol/L
	2+	30 mg/dL	3.0 mmol/L
	3+	80 mg/dL	8.0 mmol/L
GLU	-	neg.	neg.
	±	60 mg/dL	3 mmol/L
	1+	100 mg/dL	5.5 mmol/L
	2+	250 mg/dL	14 mmol/L
	3+	500 mg/dL	28 mmol/L
	>=3+	1000 mg/dL	55 mmol/L
PRO	-	neg.	neg.
	±	15 mg/dL	0.15 g/L
	1+	30 mg/dL	0.3 g/L
	2+	100 mg/dL	1.0 g/L
	3+	300 mg/dL	3 g/L
	>=3+	1000 mg/dL	10 g/L
PH	5		
	6		
	7		
	8		
	9		
NIT	-		
	+		
SG	1.005		
	1.010		
	1.015		
	1.020		
	1.025		
	1.030		
VC			0 mmol/L
			0.6 mmol/L
			1.4 mmol/L
			2.8 mmol/L
			5.6 mmol/L

Table 2.1

3 Installation Procedure

3.1 Accessory

The device is packed in a carton and should be unpacked carefully. Check the package and device for obvious damage. If you find anything damaged, contact the shipping agent or your local distributor.

The accessory list of the device is shown below.

Item	Quantity
Certificate of Conformity	1
Warranty Card	1
Operator's Manual	1
Thermal print paper roll	1
Serial communication cable	1
Power cord	1
Fuse	2

Table 3.1

3.2 Operating Environment

Like all precise electronic devices, the UA-66 must not be operated for a long time at high temperature and high humidity. In order to obtain the optimal test results, please keep the environmental temperature relatively constant within 10°C~30°C with the maximum relative humidity no higher than 80%. The platform on which the device is set should be plane and stable. If the four feet of the device cannot contact the platform evenly, adjust the retractable feet. Rotate the feet counterclockwise to raise the device, or rotate them clockwise to lower the device. The supporting table where the analyzer is placed shall be able to withstand at least 5kg of weight.

Power supply: 100-240V～, 50/60Hz, Three-wire power cord

Main supply voltage fluctuations up to ±10% of the nominal voltage.

Do not use the device in a place:

- directly exposed to sunshine.
- with flammable and explosive gases.
- with the windowed opened.
- near heating or refrigeration devices.
- near strong light source.

Warning:

To ensure personal safety and normal running of the device, ground properly the protective ground wire of the power socket and do not use a connector board connected with a 2-core wire.

The device is intended for special use and must not be disassembled without permission. The power switch, main control board, conversion board, connection wire, and feeder table insert component can only be inspected and provided by the manufacturer.

Space: Greater than the machine's size (⩾362mm(Length)×303mm(Width)×125mm(Height)), please refer to Figure 3.1.

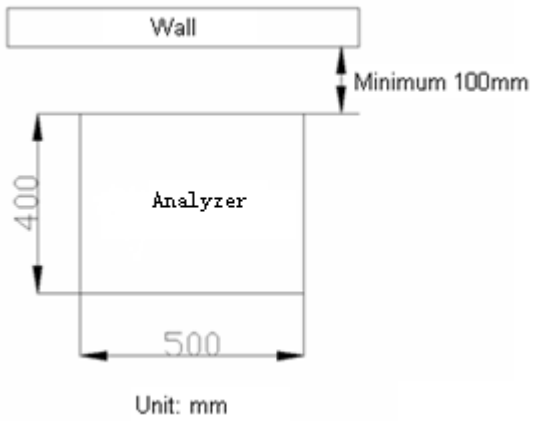

Figure 3.1

3.3 Outline and Structure of the Device

Measurement principle: The light is radiated from the LED to the strip, and reflected by the strip to the detector, and then processed by relevant circuit, thus realizing measurement of each strip.

The product mainly consists of the light source, optical receiver, feeder table insert, computer system, and display monitor.

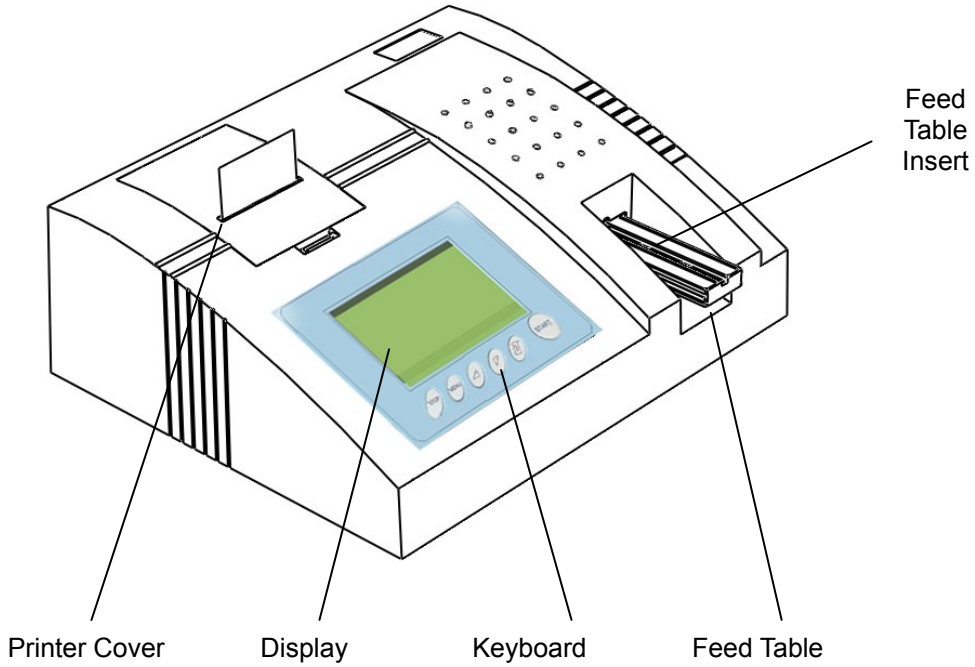

Figure 3.2

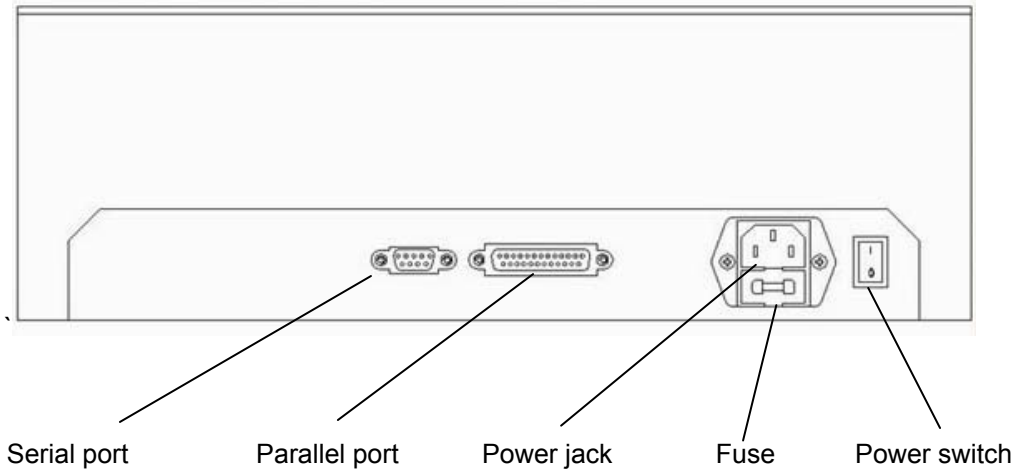

Figure 3.3

3.4 Powering On

Connect the power cord to the power outlet and toggle the power switch to the “I” position. The system software starts running and displays the following Logo.

After the power is on, the instrument shows the logo immediately on the displayer and begins to perform a self diagnosis and calibration process. Once the self diagnosis and calibration process completed, the instrument will show the system main menu including the logo, the software version, operating items, date and time.

3.5 Installation of Paper Roll

There is no paper roll in the printer when being shipped. Install a paper roll before using the instrument.

1. Be sure that the instrument is in idle state, press ESC until the display return to the main menu.

2. Lift the printer cover on the top of the instrument.

3. Remove the used paper roll core.

4. Place a new roll of paper into the printer, with the paper unrolling from the beneath.

5. Feed the paper under the roller, and to advance the paper through the printer press ▶. If the paper does not catch immediately, gently feed the paper into the roller while continuing to press ▶. To make insertion easier, be sure the leading edge is a clean, straight cut; trimming the paper to a point may also assist in easier insertion.

6. Allow 4-5 cm of paper to advance through the roller, the feed the paper through the opening on the printer cover. Place the lip of the cover under the ledge on the instrument opening and close the cover. Press down firmly on the back of the cover until it snaps into place.

Note:

When installing the print paper roll, please note that the outerface of the paper roll is the thermal face.

3.6 User Interface

Figure 2.5 shows the main menu of the instrument. The main menu is the entry of a multi-level menu system, from which all the user-selectable functions can be accessed. The main menu displays the date, time and the software version.

The main menu lists 3 menu items, “Test”, “Reports” and “Options”. The item surrounded by a shadowed box is the item being selected.

The content of the menu is the following:

Test	Single Mode	
	Slow Batch Mode	
	Fast Batch Mode	
Reports		
Options	Strip	
	Critical Value	
	Grade	
	Displayer	
	Realtime Clock	Date
		Time
	Printer	Option
		Shade
		Diagnosis
	Serial Port	9600
		19200
		38400
		57600
	Serial No	

Table 3.2

The operator executes the designated functions using the keyboard. The Keyboard consists of 3 arrow keys and 3 function keys.

Key	Function

Key	Function
FUN	Used to show or close a pop-up menu or mark a specimen during testing.
▲	Used to select the previous menu or option, or increase the selected digit while a number is being edited.
▼	Used to select the next menu or option, or decrease the selected digit while a number is being edited.
▶	Used to select the next digit while editing an integer, to feed paper at the main menu or to select the menu in the next column while multi column being displayed.
ESC	Used to stop current operation; Used to exit from current menu to upper menu level.
START	Used to begin a reagent test or to make new settings accepted;

Table 3.3

When any menu under the main menu being displayed, the menu title is shown at the top of the displayer to indicator the location of the current menu, for example:

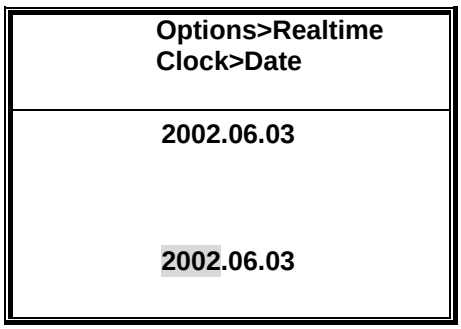

Figure 3.4

Sometimes pop-up menu is used to accept instructions from the operator. Press FUN to show a pop-up menu, for example:

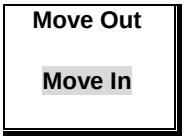

Figure 3.5

The highlighted item is the item being selected. Press START to execute the corresponding function, ESC or FUN to close the pop-up menu.

When a parameter need is to be input, the software will show an input box, for example:

Select Date
2002-05-23

Figure 3.6

If the parameter being edited has several segments, use ▶ to change the select fields and press ▲ / ▲ to modify. Press START to validate the input.

When an error occurs, the software will show a notice box with a short and a long buzz, for example:

E21
Printer Paper Empty!

Figure 3.7

When a long process being done, the software will show a notice box to indicate the status, for example:

Printing…

Figure 3.8

Usually, the upper line describes the parameter and the lower line shows the parameter to be input or modified. Input the value from the keypad and then press **START** to save the input.

3.7 Powering Off

Toggle the power switch to the “O” position to power off the device.

Before powering off the device, the user is recommended to check that there are no pending strips on the feeder table, or if any, press **ESC** to terminate the test, and then switch off the power when the strips on the feeder table are cleared.

4 Settings

At the main menu, use ▲ and ▼ as necessary to select "Options", and press START to enter "Options" menu:

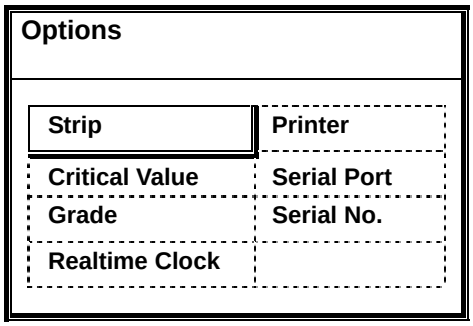

Figure 4.1

Using this menu, the operator can:

- change the date and time.
- set up the printer and serial port.
- select a strip model and set the reference ranges.
- select a grade unit.

4.1 Strip Model

At the “Options” menu, use ▲ and ▼ to select “Strip”, press START to enter the “Options>Strip” menu:.

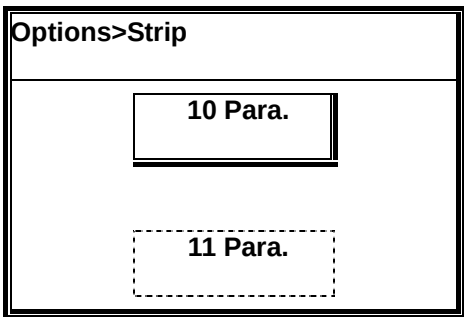

Figure 4.2

Use ▲ and ▼ as necessary to select the designated reagent strip, press START to validate the new selection.

Make sure that the strip model you have selected is consistent with the strips you are using. Inconsistent strips may lead to incorrect results.

4.2 Reference Ranges

Critical values are a set of values to be compared with the testing results. If the result of a certain parameter is the same as or higher than the corresponding critical value, a “*” will be inserted in front of the corresponding row while the report being displayed and printed.

At the “Options” menu, use ▲ and ▼ to select “Critical Value”, press START to enter the “Options>Critical Value” menu:

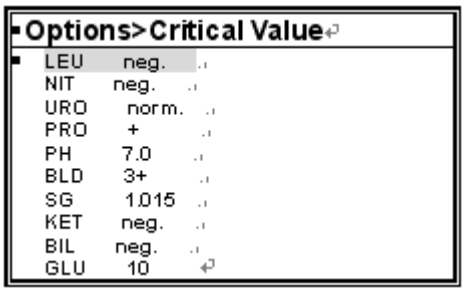

Figure 4.3

Use ▲ and ▼ to select a parameter to be modified, press START to show a pop-up window:

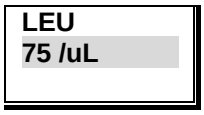

Figure 4.4

Use ▲ and ▼ to change the value, press START to validate the modification or press ESC to abandon.

4.3 Grade

The “Options>Grade” menu is used to select the grade unit system in which the results being displayed and printed. The selectable unit systems are Arbitrary, Conversional and SI. For detail, refer to Table1.1.

At the “Options” menu, use ▲ and ▼ to select “Grade”, press START to enter “Options>Grade” menu:

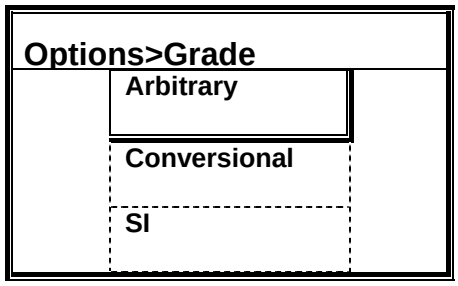

Figure 4.5

Use ▲ and ▼ as necessary to select the designated unit system, press START to validate the new selection.

4.4 Clock

The instrument equipped a realtime clock chip to afford date/time information while a report is generated. After the instrument was unpacked, the realtime clock should be corrected according to the local time immediately.

At the “Options” menu, use ▲ and ▼ to select “Realtime Clock”, press START to enter the “Options>Realtime Clock” menu:

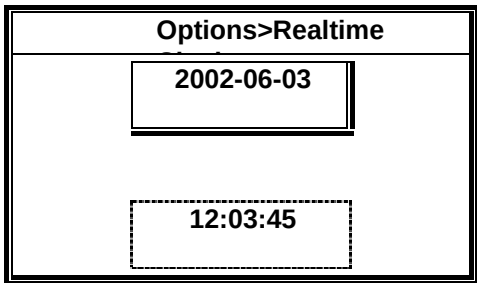

Figure 4.6

Use ▲ and ▼ to select the date, press START to modify the date:

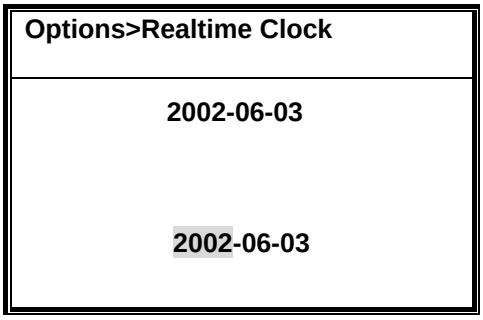

Figure 4.7

Press ▶ to move highlighted block among the year, month and day fields. Press ▲ and ▼ to modify the value of the current field. Press START to validate the new date, and press ESC to return to the “Options>Realtime Clock” menu.

At the “Options>Realtime Clock” menu, use ▲ and ▼ to select the date, press START to modify the date:

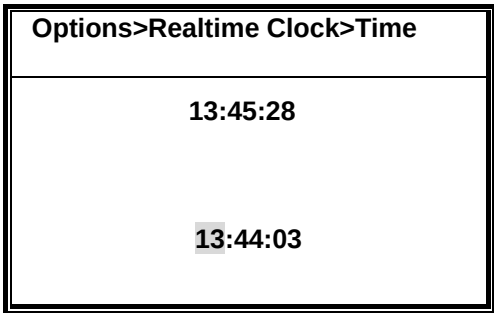

Figure 4.8

Press ▶ to move highlighted block among the hour, minute and second fields. Press ▲ and ▼ to modify the value of the current field. Press START to validate the new time, and press ESC to return to the “Options>Realtime Clock” menu.

4.5 Printer

The “Options>Printer” menu used to set auto print option and print shade or perform a test print.

Note: The device, if equipped with a serial port, can be connected with an external printer to print reports. Since printers are available in many brands and models, it is impossible that all printers can support the device and meet the requirements of print format. If the external printer cannot work normally, contact the manufacturer or the local distributor. The external printer and the internal printer output data synchronously and cannot be set up separately on the screen.

At the “Options” menu, use ▲ and ▼ to select “Printer”, press START to enter “Options>Printer” menu:

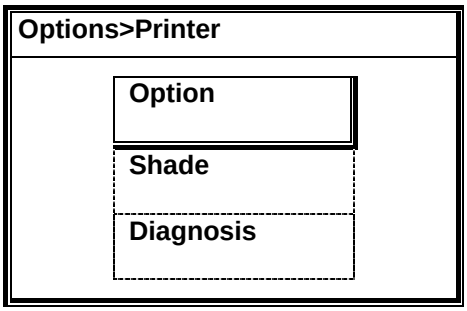

Figure 4.9

Use ▲ and ▼ to select “Auto Print”, press START to modify the auto print option:

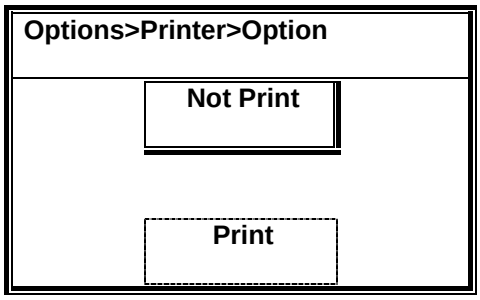

Figure 4.10

Press ▲ and ▼ to select “Yes” or “No”, press START to validate the new option. If “Yes” is selected, the instrument will print a report after a urine sample was tested.

At the “Option>Printer” menu, use ▲ and ▼ to select “Shade”, press START to modify the print shade:

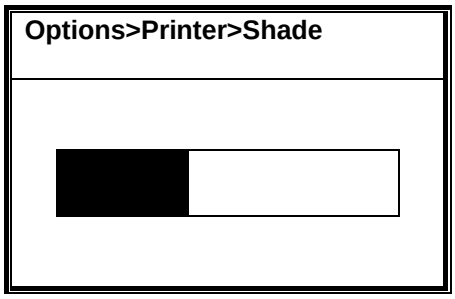

Figure 4.11

A bar shows the relative value of the print shade. Press ▲ to increase the shade and ▼ to decrease the shade. Press START to validate the new value.

The print shade varies in different seasons. In cold seasons, the print shade should be set to higher values. In warm seasons, the print shade should be set to lower values.

4.6 Serial Port

The instrument equipped a RS-232C serial port for communication with a host computer. The “Options>Serial Port” menu is used to set the baud rate of the serial port. At the “Options” menu, use ▲ and ▼ to select “Serial Port”, press START to select the baud rate:

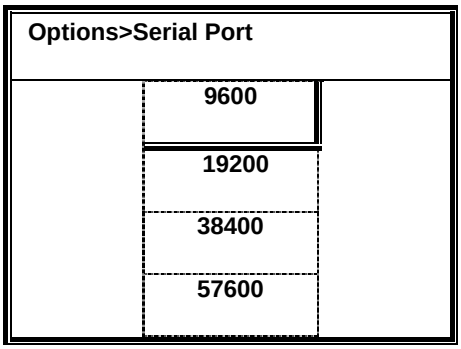

Figure 4.12

Press ▲ and ▼ to select the designated baud rate, press START to validate the new value.

4.7 Device Information

At the “Options” menu, use ▲ and ▼ to select “Serial Port”, press START to view the serial No. of the instrument:

Options>Serial No.
01234

Figure 4.13

5 Testing

The device adopts the continuous testing mode and can test up to 120 samples every hour.

Test results are managed based on sample ID, which ranges from 001 to 1000. The sample ID is numbered separately everyday and each sample ID corresponds to one test result. If a sample is tested for two times, the late result will cover the early one. Sample ID is generated automatically by the device. When powered on every day, the device automatically sets the current sample ID as 001. Every time when a sample is tested, the default sample ID increases by 1. The user is also allowed to define sample ID.

5.1 Decide Testing Mode

Three testing modes are listed on the displayer and the mode used recently is automatically selected. Using ▲ and ▼ to select the desired mode, then press START , the display goes to the next step to decide the sample No. If the mode selected by the operator differs from the recently used mode, the software will save this new setting to the Flash ROM.

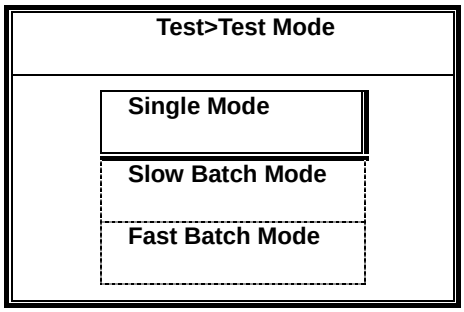

Figure 5.1

5.2 Decide Sample No.

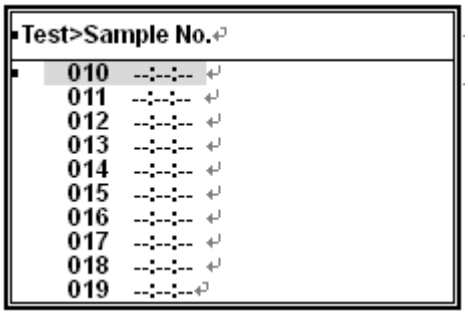

Figure 5.2

5 samples are listed on the displayer and the sample No. next to the latest sample is automatically selected. Sample No. and the time when being tested are used to identify a sample. A prefixed "*" indicates that the sample has at least a result higher than the corresponding critical value, and a prefixed " " indicates a sample marked by the operator during the testing. "--:--:--" indicates no testing was done under this No. Press ▲ and ▼ to change the selection. The list will scroll when operator try to select the sample previous to the sample listed on the top or next to the sample listed on the bottom. The operator can also enter the desired sample No. directly by pressing FUN to pop up an input box:

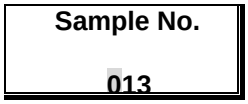

Figure 5.3

Press ▶ to change the select digit and press ▲ / ▲ to modify. Press START to validate the input.

When the sample No. selection is done, press START to start testing under the mode selected before.

5.3 The Single Mode

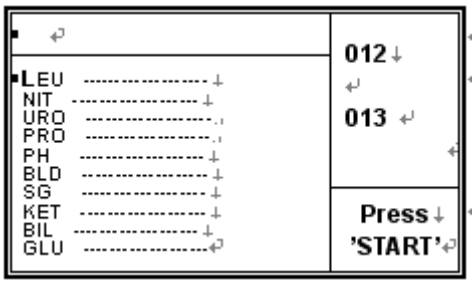

Figure 5.4

Figure 5.4 shows the display of single mode testing. The display area is divided into 4 blocks. The top-left is the sample number area, the bottom-left is the report area, the top-right is the progress area, and the bottom-right is the user prompt area.

Press START to start testing. The internal timer of the instrument begins to count down and a progress bar is shown under the sample number at the progress area. With 3 short noises from the buzzer, and the message “Dip 012” shown at the prompt area, the operator is prompted with immerse all reagent areas of the reagent strip in the urine; following the long noise from the buzzer and a “Place 012” prompt, place the reagent strip on the feed table. Then the feed table will move to the inside of the instrument. The instrument will test the stripe block by block when the feed table moving outside. When the feed table return to the original position the entire test cycle is over, and the result of the test is displayed at the report area, and output via the thermo printer.

Press START to start a new test cycle.

5.4 The Batch Modes

If “Slow Batch Mode” was selected, the instrument will prompt operator to dip the reagent strip 60 seconds after the first strip test started.

If “Fast Batch Mode” was selected, the instrument will prompt operator to dip the reagent strip 30 seconds after the first strip test started.

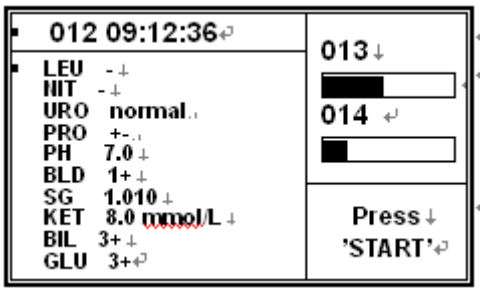

Figure 5.5

At any moment during the test cycle, operator can browse the reports generated recently using ▲ and ▼. Operator can mark a report by press FUN. The mark can be seen later in the report menu.

5.5 Stop Testing

At any moment during the test cycle, operator can stop the testing process by pressing ESC, the display will return to the main menu.

5.6 Testing Procedures

Take a strip and dip the whole strip into a test tube of urine sample. When the strip is soaked, take it out carefully to avoid carrying out excessive urine, and then use absorbent paper to absorb extra urine on the strip.

At the Single Mode or Batch Modes, pressing START , When a “beep” is heard, Use a hand to place the prepared strip properly on the feed table. It indicates the test will start.

Note:

The reagent area of the strip must face up.

Each strip must be placed vertically between the first column of bulge carriers and the second column of bulge carriers on the feed table.

The far tip (the tip that is not in hand) must exceed the top of the bulge carriers.

5.7 Testing Precautions

- Make sure that the strips for testing match the strip-type settings of the analyzer and are still within the valid period; otherwise incorrect results may occur.
- After soaked and dried, the strip should be placed on the feed table immediately to avoid padding the reaction time and affecting the result accuracy.

- Try to absorb all extra urine from the strip to avoid cross-contamination on the feed table due to too much extra urine accumulated.
- Do not pull the feed table or remove the waste box; otherwise the analyzer will not function correctly.
- Wash the feed table and empty the waste box everyday.

It is unnecessary to exit the testing window or shut down the analyzer when there is no sample to test temporarily.

6 Report Query

6.1 Querying Reports by Date

At the main menu, use ▲ and ▼ to select “Reports” and press START, the instrument begins to search through the report storage, then lists all the date in ascending order. The most recent date is listed at the bottom of the display.

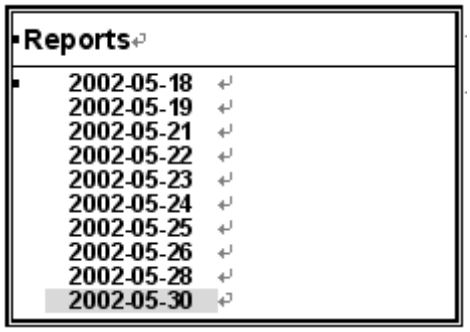

Figure 6.1

Use ▲ to select the previous date and ▼ to select the next date. At this moment, the operator can

- ♦ Change the selected date by input a date directly.
- ♦ Delete the reports generated between two specified dates;
- ♦ Transmit reports of the selected date;

♦ Print reports of the selected date;

Press FUN to show a pop-up menu.

Select Date
Delete
Transmit
Print

Figure 6.2

Use ▲ and ▼ to select “Select Date” and press START, the instrument prompts the operator to input a new date:

Select Date
2002-05-23

Figure 6.3

Use ▶ to move the high lightened block among the year, month and day fields, use ▲ and ▼ to modify the highlighted field and press START when done. The date list in Figure 6.1 will scroll to the specified date. If the specified date does not exist in the list, The date list will scroll to the nearest date.

Select “Delete” item of the pop-up menu (See Figure 6.2) and press START to delete reports generated between 2 specified dates. First the instrument asks the operator to enter the start date.

From
2002-05-23

Figure 6.4

Use ▶ to move the high lightened block among the year, month and day fields, use ▲ and ▼ to modify the highlighted field and press START when done. Then the instrument asks the operator to enter the end date.

To
2002-05-23

Figure 6.5

Use ▶ to move the high lightened block among the year, month and day fields, use ▲ and ▼ to modify the highlighted field and press START when done. At last the instrument prompts the operator to confirm.

Delete the reports
from 2002-05-19

to 2002-05-23 ?

Yes No

Figure 6.6

Select “Yes” and press START to delete the reports. Select “No” and press START to abandon.

Select “Transmit” item of the popup menu (See Figure 6.2) and press START to transmit reports of the selected date. First the instrument asks the operator to enter the start sample No.

Start
No.
000

Figure 6.7

Modify the start sample No. using ▲, ▼ and ▶, and then press START when done. The instrument asks the operator to enter the end sample No.

End
No.
999

Figure 6.8

Modify the end sample No. using ▲, ▼ and ▶, and then press START when done. The instrument begins to transmit the reports between the start No. and the end No. to the host computer via RS-232C serial port. A message box will be shown during the transmission.

Transmitting…

Figure 6.9

Press ESC to stop the transmission.

Select “Print” item of the pop-up menu (see Figure 6.2) and press START to print reports of the selected date. Just as the transmitting process described above, the instrument will ask the operator to enter the start No. and the end No. Press START

to print the reports between the start No. and the end No. when the end number was entered. A message box will be shown during the printing process.

Printing…

Figure 6.10

Press ESC to stop the printing process.

6.2 Recalling Reports

On the date list (See Figure 6.1), press START to browse the reports of the specified date.

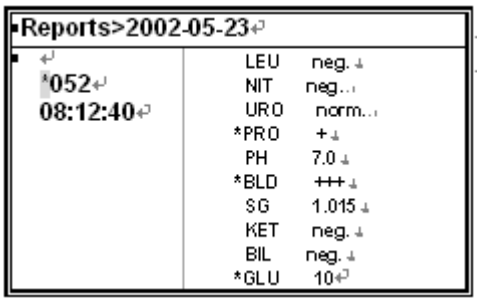

Figure 6.11

If the date is today, the report of the most recently testing will be first displayed, otherwise the report with the smallest sample No. will be first displayed. Press ▲ to view the previous report and ▼ to view the next report. Press FUN to show a pop-up menu.

Mark

Select No.

Transmit

Print

Figure 6.12

Note that the “Mark” item will be shown only if the report of today is shown. Use ▲ and ▼ to select the designed item, press START to perform the corresponding function.

Select “Print” and press START to print the report.

Select “Transmit” and press START to print to the host computer via the RS-232C serial port.

Select “Mark” and press START to mark the report. The mark can be seen at the “Testing>Sample No” menu. If the report being currently displayed has already been marked, “Remove Mark” will be shown instead of “Mark”.

Select “Select No.”, the instrument prompt the operator to enter a new sample No..

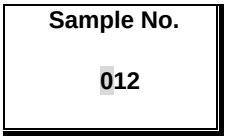

Figure 6.13

Modify the end sample No. using ▲, ▼ and ►, and then press START when done. The report with the specified sample No. will be shown if exists, otherwise the report with the nearest sample No. will be shown.

7 Troubleshooting

When some or all functions of the analyzer cannot work normally due to system failure or misoperation by the operator, an error message will be displayed. The following table shows all error messages, including error code, error description, and solutions.

Error ID	Message	Solution
E01	System Memory failure	The system memory failed and the instrument can't work. Please contact the customer service.
E02	System Data Failure	The calibration data lost and the instrument can't work. Please contact the customer service.
E03	Feed Table Failure	The instrument can't find the mechanical origin of the feed table. Please turn off the power off and retry for few more times. If not succeeded, please contact the customer service.
E04	Calibration Error	Clean the white plastic calibration chip as instructed in 7.2 Daily Cleaning, and turn the instrument once more. If not succeeded, contact with the customer service.
E05	Feed Table Polluted	If any foreign material is in the reagent strip holding channel, remove it; If the feed table insert is heavily polluted by the urine specimen, clean it as instructed in 7.2 Daily Cleaning.

Error ID	Message	Solution
E10	Stray Light Error	Confirm if the instrument is located under direct sunlight, or there is a strong light source working near the instrument. Please place the instrument as instructed in 3.2 Operating Environment.
E11	Positioning Error	The urine sample being tested should be tested once more. If this error usually occurs, contact the customer service.
E12	Strip Placement Error	Confirm if a reagent strip is placed on the feed table and the tip of the strip touches the end of the strip holding channel. If not, replace the strip in 10 seconds and press START.
E21	Printer Paper Empty	Please install a new paper roll.
E22	Printer Handle	Put the handle of the printer to the “down” position.

Table 7.1

8 Maintenance

8.1 Daily Maintenance and Tips

1. While the analyzer is performing tests, do not expose it to direct sunshine to avoid influencing the test precision.

2. Often use soft and dry cloth to clean the analyzer. If dirt is found on the surface of the analyzer, clear it with fresh water. Do not use gasoline, diluted paint, benzene compounds, alcohol and other organic solvents, which may cause the analyzer to deform or fade, thus influencing the system performance and appearance.

3. Use clear soft dry cloth or soft paper instead of water to clean the LCD screen.

4. Keep the feed table clean. If urine is accumulated during testing, wipe it with absorbent paper to avoid influencing the test results due to cross-contamination. Clean the feed table with fresh water every day.

5. This product can be used only by qualified clinical personnel.

Note:

In view of the properties of strips, do not use it at too high or too low temperature to ensure accurate results. The ideal operating temperature is within 10°C ~30°C.

Only use strips that match the urine analyzer. Use the controls recommended by our company to ensure consistent and accurate results. Using Bio-Rad Laboratories,Irvine,CA : Urinalysis Control level 1 and level 2.

Disposal of waste strips is subjected to waste disposal regulations. Please reject waste strips according to the local waste disposal regulations.

Use fresh sample (collected for 2-3 hours) for testing.

The following table lists the tools that may be required in the maintenance operations.

No.	Tools
1.	Medical gloves
2.	Distilled water
3.	Cotton cloth
4.	Absorbent paper
5.	Disinfectant (see 8.3 for details)

8.2 Disassembling and Cleaning Feed Table

The feed table and feed table insert must be kept clean to obtain the best result with the UA-66 instrument / reagent strip system. At the end of each daily use, the instrument‘s white plastic calibration chip, located on the feed table insert, and the reagent strip holding channel should be cleaned. To clean the feed table insert, perform the followings:

1. Turn the instrument on. The feed table will move out from the “storage” position to the “load” position.

2. Grasp the front end of the insert on each side with the thumb and one finger. Apply a counter pressure with the palm of the hand against the feed table and gently pull the insert outward. After the insert is completely exposed, lift the front end of insert up and off the table.

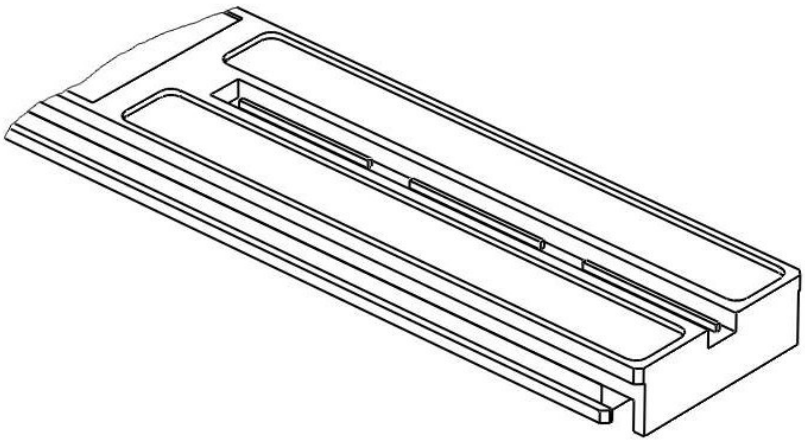

Figure 8.1

3. Clean the insert using a cotton-tipped applicator stick or other suitable absorbent material wetted with distilled water. Thoroughly swab the strip holding channel area until it is clear. Inspect the surface of the white plastic calibration chip for dust, foreign material, scratches or scuffs. If necessary, gently wipe the calibration chip with a cotton-tipped applicator stick wetted with distilled water. Allow the calibration chip to air dry before using. Make sure no foreign particles remain on its surface. If the calibration chip cannot be cleaned or is scratched or scuffed, contact the customer service for replacement.

Note:

Do not use any material that will scratch the feed table insert or the white plastic calibration chip. Do not use solvents of any kind to clean the calibration chip.

4. Wipe the feed table surface and feed table mounting groove with a damp cloth. Wipe the feed table dry with an absorbent tissue.

5. To replace the feed table insert in the feed table:
Hold the feed table insert in the same manner as when it was removed(holding it by the end opposite the calibration chip). With the front end angled upwards, insert the calibration chip end about midway onto the feed table. Slide the insert in while lowering the front end until the insert is laying flat against the table and resistance is felt. Then place the thumb against the front edge of the insert and two fingers against the underside of the feed table. Apply a counter-pressure to the feed table while pressing the insert firmly inward. The insert is properly positioned in the feed table when the front edge is flush with the retain rails of the feed table.

6. After replacing the feed table insert, the LEDs must be recalibrated. To ensure this occurs, turn the instrument off, wait several seconds, and then turn it on again before testing additional samples.

Biohazard:

Take necessary protection measures during operation.

Caution:

The feed table is an important part of the device and must be kept properly. Do not use detergent or hot water to clean the feed table or bake or heat it. A deformed feed table may directly influence the test results.

8.3 Disinfection

The only part of the UA-66 instrument that comes into contact with the urine specimen is the feed table insert. To disinfection the insert, perform the followings:

1. Remove, clean and dry the feed table insert as instructed in 8.2.

2. Prepare one of the following solutions for use in disinfection:

   <a>. 2% glutaraldehyde solution - Cidex™ (use full strength) is a commercially available product that is acceptable for use. Refer to the labeling accompanying this disinfection product for complete instructions on its use.

   <b>. 0.05% sodium hypochlorite solution – dilute 1 mL of 5% sodium hypochlorite (e.g., Clorox® Bleach) into 99 mL of water for a 1:100 dilution.

   <c>. Isopropyl alcohol (70% - 85%), undiluted.

3. Fill a tall, narrow container to a depth of 4 inches (10 cm) with the disinfection solution.

4. Immerse the feed table insert into the disinfection solution. So the white plastic calibration chip must be above the liquid level.

5. Allow the feed table insert remain in the solution for 10 minutes.

6. Remove the feed table insert and rinse thoroughly with distilled water to remove any residual solution.

7. Wipe the feed table insert in dry with an absorbent tissue.

8. Replace the feed table insert in the feed table, as instructed in 8.2 Daily Cleaning.

Biohazard:

Take necessary protection measures during operation.

8.4 Replacement of Fuse

Disconnect the power cord, use a slotted screwdriver to open the fuse box, and then remove the fuse. If the fuse is burnt, replace it with a new one. Close the fuse box.

Note:

Use the fuse provided with the analyzer or a slow fuse (tube): 5mm wide, 20mm long, 250V/2A, T2AL250V.

8.5 Periodic Cleaning

If the feed table insert becomes excessively dirty and cannot be adequately cleaned as described in 8.2 Daily Cleaning, the following procedure should be used:

1. Remove the feed table insert from the table, as described in 8.2 Daily Cleaning.

2. Wipe the strip holding channel of the feed table insert with 0.1N sodium hyndroxide (NaOH) using a cotton-tipped applicator stick or other suitable absorbent material. Thoroughly swab the channel area until it is clean.

3. Rinse the feed table insert channel thoroughly with water to remove any residual NaOH.

4. Wipe the feed table surface and feet table mounting groove with a damp cloth. Wipe the feed table dry with an absorbent tissue.

5. Replace the feed table insert in the feed table, as instructed in 8.2 Daily Cleaning.

9 Serial Communication

9.1 Cable Connection

The serial cable provided with the analyzer is used to connect a computer with the RS-232 interface on the analyzer. On one end of the cable are 9 pins and on the other end are 9 holes. Refer to the table below for the cable connection.

DB9 Male on the Analyzer	DB9 Female on the Computer
2	2
3	3
5	5

Table 9.1

9.2 Basic Specifications

Parameter	Specification
Communication mode:	Asynchronous serial communication
Signals used:	TXD, RXD, and GND
Baud rate:	9600, 19200, 38400, and 57600bps
Parity:	Odd

Parameter	Specification
Data length:	8 Bits
Stop digit:	1 Bit

Table 9.2

9.3 Transmission Content

When the analyzer sends data to a computer, the analyzer is as the sending end and just sends instead of receiving something; the computer is as the receiving end, and just receives without replying.

Data content uses frame as its unit, and one frame data is a report. A frame is composed of a number of lines, and ends with a line containing ODH and OAH. A line is composed of fields, which are separated by 09H.

The table below describes the details of a frame.

Line	Field 1	Field 2	Description
1	Date, 20XX-XX-XX		Only one field
2	Time, XX:XX:XX		Only one field
3	Sample ID, XXX		Only one field
4			Blank line
5	Name of test item 1	Result of test item 1	
6	Name of test item 2	Result of test item 2	
7	Name of test item 3	Result of test item 3	
8	Name of test item 4	Result of test item 4	
9	Name of test item 5	Result of test item 5	
10	Name of test item 6	Result of test item 6	
11	Name of test item 7	Result of test item 7	
12	Name of test item 8	Result of test item 8	
13	Name of test item 9	Result of test item 9	
14	Name of test item 10	Result of test item 10	
15	Name of test item 11	Result of test item 11	This line will be blank in the case of 10 test items.

Table 9.3

If the test results contain “±” and “μ”, the two symbols will be replaced by “+-” and “u”.

After receiving a frame of data, the computer will give no replies.